Light

CPU Architecture & Von Neumann

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AQA GCSE Computer Science — **Computer Systems: CPU Architecture**. This topic covers the internal components of the Central Processing Unit, how instructions are processed through the Fetch-Execute cycle, and the factors that affect CPU performance. This content is assessed in **Paper 2 (Component 2): Computing concepts**.

Exam questions typically ask you to **explain** the role of CPU components, **describe** the stages of the Fetch-Execute cycle, and **explain** how factors such as clock speed, cache size, and number of cores affect performance. Expect questions worth 2—6 marks requiring you to demonstrate understanding of how these components work together.

Why CPU architecture matters

The Central Processing Unit (CPU) is the brain of a computer system. It processes all instructions and performs calculations that make software run. Understanding CPU architecture helps you:

- **Understand how computers execute programs:** Every line of code you write is ultimately processed by the CPU through a series of fetch, decode, and execute operations

- **Make informed hardware decisions:** Knowing what clock speed, cores, and cache do helps when comparing or specifying computer systems
- **Debug performance issues:** Understanding the CPU helps explain why some programs run slowly or why certain operations are faster than others
- **Connect software to hardware:** Programming concepts like variables and loops translate into physical operations within the CPU

The AQA specification requires you to understand the major CPU components, how they interact through buses, and how the Fetch-Execute cycle processes instructions continuously.

What you must know

This checklist covers the key concepts for CPU architecture. You should be able to explain each point clearly.

- **Explain the role and operation** of the Arithmetic Logic Unit (ALU).
- **Explain the role and operation** of the Control Unit (CU).
- **Explain the role and operation** of the clock.
- **Explain the role and operation** of registers.
- **Explain the role and operation** of buses.
- **Explain the role and operation** of main memory in relation to the CPU.
- **Understand and explain** the Fetch-Execute cycle, including what happens at each stage.
- **Explain the effect** of clock speed on CPU performance.
- **Explain the effect** of the number of processor cores on CPU performance.
- **Explain the effect** of cache size on CPU performance.

> **Specification note.** AQA states that knowledge of specific registers is not required. You need to understand that registers exist and their general purpose, but you will not be asked to name or describe individual registers. Focus on the general role of registers as small, fast, temporary storage locations within the CPU.

Toolkit

CPU components at a glance

Component	Role	Key point for exams
Arithmetic Logic Unit (ALU)	Performs arithmetic and logical operations	Calculations (+, −, ×, ÷) and comparisons (<, >, =)
Control Unit (CU)	Manages and coordinates CPU operations	Fetches, decodes instructions; sends control signals
Clock	Synchronises CPU operations	Sends regular pulses; speed measured in Hz
Registers	Fast, temporary storage within CPU	Hold data/instructions currently being processed

Component	Role	Key point for exams
Buses	Wires that transmit data/signals between components	Three types: data bus, address bus, control bus

The Fetch-Execute cycle summary

Stage	What happens
Fetch	The next instruction is fetched from main memory to the CPU
Decode	The instruction is decoded to work out what it is
Execute	The instruction is executed (carried out); this may include reading/writing from/to main memory

This cycle repeats continuously while the computer is running. The CPU fetches one instruction, decodes it, executes it, then fetches the next instruction, and so on.

Performance factors summary

Factor	Effect on performance	Typical values
Clock speed	Higher clock speed = more cycles per second = faster processing	2 GHz — 5 GHz
Number of cores	More cores = more instructions processed simultaneously	2, 4, 6, 8, or more cores
Cache size	Larger cache = more data stored close to CPU = fewer slow memory accesses	4 MB — 64 MB

CPU component diagram

This diagram shows the main components of the CPU and how they connect to main memory via buses:

```
CONTROL UNIT  ◄──────────►  ALU

• Fetches                   • Arithmetic
• Decodes                   • Logical
• Coordinates               • Comparisons

                 ▼
            REGISTERS  ◄── CLOCK (timing)
          (fast storage)

ADDRESS BUS ↓   DATA BUS ↕   CONTROL BUS ↕

                 ▼
            MAIN MEMORY
               (RAM)

          Stores programs
          and data
```

Key: ↓ = unidirectional (one way), ↕ = bidirectional (both ways)

> **Exam tip:** When explaining performance factors, always link cause to effect. For example: "A higher clock speed means more Fetch-Execute cycles can be completed per second, so more instructions are processed in the same time, making the CPU faster."

1. CPU components

The Central Processing Unit (CPU) is the part of the computer that processes instructions and data. It contains several key components that work together to execute programs. The AQA specification requires you to understand the role and operation of each component.

1.1 The Arithmetic Logic Unit (ALU)

The **Arithmetic Logic Unit (ALU)** is the component of the CPU that performs all arithmetic and logical operations.

Arithmetic operations include:

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)

Logical operations include:

- Comparisons: equal to, not equal to, less than, greater than
- Boolean logic: AND, OR, NOT
- Binary shifts

When your program performs a calculation like `total = price * quantity`, the ALU carries out the multiplication. When your program checks a condition like `if score > 50`, the ALU performs the comparison.

The ALU receives data from registers, performs the required operation, and stores the result back in a register. It works under the direction of the Control Unit, which tells it what operation to perform.

1.2 The Control Unit (CU)

The **Control Unit (CU)** is the component that manages and coordinates all CPU operations. It acts as the manager of the CPU, directing the flow of data and instructions.

The Control Unit's responsibilities:

- Fetches instructions from main memory
- Decodes instructions to determine what action is needed
- Sends control signals to other components to coordinate execution
- Controls the timing of operations using signals from the clock
- Manages the flow of data between the CPU, memory, and input/output devices

The Control Unit does not perform calculations itself — that is the ALU's job. Instead, the CU orchestrates when and how the ALU, registers, and memory interact to execute each instruction.

1.3 The Clock

The **clock** is a component that generates a regular electrical signal (pulse) to synchronise all CPU operations. Every operation within the CPU happens in step with these clock pulses.

Key points about the clock:

- Produces a regular timing signal measured in Hertz (Hz)
- 1 Hz = 1 cycle per second
- Modern CPUs operate in gigahertz (GHz): 1 GHz = 1 billion cycles per second
- Each clock cycle, the CPU can perform a basic operation

Think of the clock like a conductor keeping an orchestra in time. Without synchronisation, different parts of the CPU would operate at different speeds, causing chaos. The clock ensures all components work together in a coordinated manner.

1.4 Registers

Registers are small, extremely fast storage locations within the CPU. They hold data and instructions that are currently being processed.

Key characteristics of registers:

- Very small capacity (typically 32 or 64 bits each)
- Extremely fast access — faster than cache or RAM
- Located inside the CPU itself
- Hold data temporarily during processing

Registers are essential because the CPU cannot operate directly on data in main memory. Data must first be loaded into registers, processed by the ALU, and results stored back to registers before being written to memory.

The AQA specification states that knowledge of specific registers is not required. You need to understand that registers exist and provide fast temporary storage, but you will not be asked about individual named registers.

> **Exam tip:** When describing registers, emphasise that they are the fastest form of storage and are located within the CPU itself. This distinguishes them from cache (fast but outside the CPU core) and RAM (slower, in main memory).

1.5 Buses

A **bus** is a collection of wires through which data and signals are transmitted from one component to another. Buses connect the CPU to main memory and other components, allowing them to communicate.

There are three main types of bus in a computer system:

Data Bus

Carries the actual data being transferred between components.

- **Bidirectional:** data flows in both directions
- Transfers data from memory to CPU (read)
- Transfers data from CPU to memory (write)

Address Bus

Carries memory addresses that identify where data should be read from or written to.

- **Unidirectional:** signals flow from CPU to memory only
- The CPU specifies which memory location to access

Control Bus

Carries control signals that coordinate and manage operations.

- **Bidirectional:** signals flow both ways
- Signals include: read, write, clock, interrupt

When the CPU needs to read data from memory, it places the memory address on the address bus, sends a "read" signal on the control bus, and receives the data via the data bus. When writing, the CPU places the address on the address bus, the data on the data bus, and sends a "write" signal on the control bus.

1.6 Main memory and the CPU

Main memory (RAM) stores the programs and data that the CPU is currently working with. The CPU cannot execute instructions directly from secondary storage — programs must be loaded into main memory first.

The relationship between the CPU and main memory is fundamental:

- Instructions are stored in main memory at specific addresses
- The CPU fetches instructions from main memory one at a time
- Data needed by instructions is read from main memory into registers
- Results are written from registers back to main memory

The buses provide the communication pathway between the CPU and main memory. Without this connection, the CPU would have no instructions to execute and no data to process.

Worked Example 1 – Explaining the ALU

Question: Explain the role of the Arithmetic Logic Unit (ALU) in a CPU. [2 marks]

Model answer:
The ALU performs arithmetic operations such as addition, subtraction, multiplication and division **(1)**. It also performs logical operations such as comparisons (greater than, less than, equal to) and Boolean operations (AND, OR, NOT) **(1)**.

Exam tip: For 2 marks, give one point about arithmetic operations and one about logical operations. Name specific examples to demonstrate understanding.

Worked Example 2 – Explaining the Control Unit

Question: Describe the role of the Control Unit in the CPU. [3 marks]

Model answer:
The Control Unit fetches instructions from main memory **(1)**. It decodes instructions to determine what operation needs to be performed **(1)**. It sends control signals to coordinate the actions of other CPU components such as the ALU and registers **(1)**.

Exam tip: Link the Control Unit to the Fetch-Execute cycle stages — it is responsible for the fetch and decode stages, and coordinates the execute stage.

Worked Example 3 — Explaining buses

Question: Explain the purpose of the address bus and the data bus. [4 marks]

Model answer:
Address bus: The address bus carries memory addresses from the CPU to memory **(1)**. It identifies the location in memory where data should be read from or written to **(1)**.
Data bus: The data bus carries the actual data being transferred between the CPU and memory **(1)**. It is bidirectional, meaning data can flow from memory to CPU (read) or from CPU to memory (write) **(1)**.

Exam tip: When comparing buses, mention the direction of data flow — address bus is unidirectional (CPU to memory), data bus is bidirectional.

Worked Example 4 — Explaining registers

Question: Explain what registers are and why they are needed in a CPU. [3 marks]

Model answer:
Registers are small, fast storage locations within the CPU **(1)**. They temporarily hold data and instructions that are currently being processed **(1)**. They are needed because the CPU cannot operate directly on data in main memory — data must be loaded into registers first for the ALU to process **(1)**.

Exam tip: Emphasise speed and location (inside the CPU) as key characteristics that distinguish registers from other types of memory.

2. The Fetch-Execute cycle

The **Fetch-Execute cycle** (also called the Fetch-Decode-Execute cycle or instruction cycle) is the basic operation cycle of the CPU. It describes how the CPU retrieves instructions from memory, interprets them, and carries them out. This cycle repeats continuously while the computer is running.

The three stages explained

Stage 1: Fetch

Stage 2: Decode

The next instruction is fetched from main memory to the CPU.

- The CPU checks which instruction to fetch next
- The memory address is sent via the address bus
- The instruction is retrieved via the data bus
- The instruction is stored in a register in the CPU

The instruction is decoded to work out what it is.

- The Control Unit examines the instruction
- It determines what operation is required
- It identifies what data or memory locations are needed
- It prepares the necessary components for execution

Stage 3: Execute

The instruction is executed (carried out).

- The required operation is performed
- This may involve the ALU for calculations
- This may include reading from main memory
- This may include writing to main memory

The Fetch-Execute cycle diagram

The cycle repeats continuously while the computer is running:

```
     ┌──────────────────────────────────────────────────┐
     │                                                  │
     ▼                                                  │
┌─────────┐         ┌─────────┐         ┌─────────┐     │
│         │         │         │         │         │     │
│  FETCH  │────────►│ DECODE  │────────►│ EXECUTE │─────┘
│         │         │         │         │         │
└─────────┘         └─────────┘         └─────────┘
     │                   │                   │
     ▼                   ▼                   ▼
Get next            Work out            Carry out
instruction         what the            the
from main           instruction         instruction
memory              means               (may use ALU)
```

After the execute stage completes, the cycle immediately begins again with fetching the next instruction. This continues as long as the computer is running. A modern CPU operating at 3 GHz completes approximately 3 billion of these cycles every second.

2.1 How the components work together

The Fetch-Execute cycle demonstrates how all CPU components collaborate:

- **Control Unit:** Manages the entire cycle, directing when each stage occurs
- **Registers:** Store the current instruction and any data being processed
- **ALU:** Performs any calculations or comparisons during the execute stage
- **Buses:** Carry addresses, data, and control signals between CPU and memory
- **Clock:** Synchronises each step, ensuring operations happen in the correct order
- **Main memory:** Stores the instructions and data that the CPU fetches and writes

> **AQA specification wording.** The specification states the execute stage "may include reading/writing from/to main memory." This is important — not every instruction requires memory access during execution. Some instructions operate only on data already in registers.

2.2 A practical example

Consider the simple instruction to add two numbers and store the result. Here is how the Fetch-Execute cycle would process this:

Stage	Action	Components involved
Fetch	The CPU retrieves the "ADD" instruction from memory address 1000	Control Unit, address bus, data bus, registers
Decode	The Control Unit determines this is an addition operation and identifies the data locations	Control Unit
Execute	The ALU adds the two values; the result is stored in a register	ALU, registers

The cycle then repeats for the next instruction, which might store the result to main memory.

2.3 The continuous nature of the cycle

The Fetch-Execute cycle is continuous — it runs constantly while the computer is powered on and executing programs. Even when a computer appears idle, the CPU is still cycling through instructions (often just waiting or performing background tasks). Each program you run consists of thousands or millions of instructions. The CPU processes these one at a time (or several at a time with multiple cores), each going through the same fetch, decode, execute sequence.

Worked Example 5 – Describing the Fetch-Execute cycle

Question: Describe what happens during each stage of the Fetch-Execute cycle. [6 marks]

Model answer:

Fetch: The next instruction is fetched from main memory **(1)**. The instruction is transferred to the CPU via the data bus and stored in a register **(1)**.
Decode: The Control Unit decodes the instruction **(1)**. It determines what operation needs to be performed and what data is required **(1)**.
Execute: The instruction is carried out/executed **(1)**. This may involve the ALU performing a calculation, or reading/writing data from/to main memory **(1)**.

Exam tip: For extended answers about the FE cycle, structure your answer clearly by stage. Use the exact AQA wording: "fetched from main memory," "decoded to work out what it is," "executed (carried out)."

Worked Example 6 — Role of the Control Unit in the cycle

Question: Explain the role of the Control Unit during the Fetch-Execute cycle. [3 marks]

Model answer:
During the **fetch** stage, the Control Unit sends the memory address of the next instruction to main memory via the address bus **(1)**. During the **decode** stage, the Control Unit interprets the instruction to determine what operation is needed **(1)**. During the **execute** stage, the Control Unit sends control signals to coordinate the other components, such as directing the ALU to perform a calculation **(1)**.

Exam tip: The Control Unit is involved in all three stages. Show this breadth of involvement for maximum marks.

Worked Example 7 — Buses in the Fetch-Execute cycle

Question: Explain how the buses are used during the fetch stage of the Fetch-Execute cycle. [3 marks]

Model answer:
The **address bus** carries the memory address of the instruction to be fetched from the CPU to main memory **(1)**. The **control bus** carries a read signal to indicate that data should be read from memory **(1)**. The **data bus** carries the instruction from main memory back to the CPU **(1)**.

Exam tip: When explaining bus usage, always specify the direction of data flow and what type of information is being carried.

Worked Example 8 – Why the cycle repeats

Question: Explain why the Fetch-Execute cycle runs continuously while a computer is operating. [2 marks]

Model answer:
Programs consist of many instructions that must be executed in sequence **(1)**. The CPU must continuously fetch, decode, and execute each instruction one after another to run the program **(1)**.

Exam tip: Link the continuous cycle to the practical reality that programs contain many instructions, not just one.

3. Performance factors

The performance of a CPU — how quickly it can process instructions — depends on several factors. The AQA specification requires you to understand and explain the effect of three key factors: clock speed, number of cores, and cache size.

3.1 Clock speed

Clock speed refers to the number of cycles the CPU can perform per second, measured in Hertz (Hz). Modern CPUs operate in gigahertz (GHz), where 1 GHz = 1 billion cycles per second.

Effect on performance: A higher clock speed means more Fetch-Execute cycles can be completed per second. This means more instructions are processed in the same amount of time, making the CPU faster.

Clock speed	Cycles per second	Performance implication
2 GHz	2 billion	Suitable for basic tasks
3 GHz	3 billion	Good for general use
4 GHz	4 billion	High performance for demanding applications
5 GHz	5 billion	Very high performance, often in gaming/workstation CPUs

However, clock speed is not the only factor. A CPU with a higher clock speed will not always outperform one with a lower clock speed if other factors (such as cores or cache) are significantly different.

Important limitation: Increasing clock speed also increases heat generation and power consumption. There are practical limits to how fast a CPU can run before it overheats. This is why manufacturers focus on other performance factors as well.

3.2 Number of processor cores

A **core** is an independent processing unit within the CPU. Each core can execute its own Fetch-Execute cycle independently. A CPU with multiple cores is called a multi-core processor.

Effect on performance: More cores allow the CPU to process multiple instructions simultaneously. A dual-core processor can execute two instruction streams at once; a quad-core processor can handle four.

Term	Number of cores	Typical use
Single-core	1	Legacy systems, basic embedded devices
Dual-core	2	Basic computing, older laptops
Quad-core	4	General computing, mainstream laptops
Hexa-core	6	Gaming, content creation
Octa-core	8	High-performance workstations, gaming

Important consideration: Software must be designed to use multiple cores (called parallel processing) to benefit from them. A program that can only use one core will not run faster on an 8-core CPU than on a single-core CPU with the same clock speed.

3.3 Cache size

Cache is a small amount of very fast memory located close to (or within) the CPU. It stores frequently used data and instructions so they can be accessed more quickly than fetching them from main memory (RAM).

Effect on performance: A larger cache can store more data close to the CPU, reducing the number of times the CPU must wait for data from slower main memory. This speeds up processing because cache is much faster than RAM.

The memory speed hierarchy

Different types of memory have different speeds. Data access times increase as you move away from the CPU:

1. **Registers** — fastest (inside CPU)
2. **Cache** — very fast (close to CPU)
3. **RAM (main memory)** — fast
4. **Secondary storage (SSD/HDD)** — slowest

The cache acts as a buffer between the very fast registers and the slower RAM, reducing the performance gap.

When the CPU needs data, it first checks the cache. If the data is stored in cache, it can be retrieved very quickly. If the data is not in cache, the CPU must fetch it from slower main memory, which takes longer. A larger cache means more data can be stored close to the CPU, reducing the number of slow memory accesses needed.

3.4 Comparing performance factors

Factor	Higher value means...	Best for...
Clock speed	More cycles per second, faster single-thread performance	Tasks that run on one core, gaming, single-threaded applications
Number of cores	More tasks processed simultaneously	Multi-tasking, video editing, running multiple applications
Cache size	Fewer slow memory accesses	Applications that repeatedly access the same data

Worked Example 9 – Explaining clock speed

Question: Explain how clock speed affects the performance of a CPU. [2 marks]

Model answer:
Clock speed determines how many Fetch-Execute cycles the CPU can complete per second **(1)**. A higher clock speed means more instructions can be processed per second, making the CPU faster **(1)**.

Exam tip: Always link clock speed to the Fetch-Execute cycle — this shows you understand how clock speed actually affects processing.

Worked Example 10 – Explaining multiple cores

Question: Explain how having multiple cores can improve CPU performance. [3 marks]

Model answer:
Each core can execute its own Fetch-Execute cycle independently **(1)**. This allows multiple instructions or tasks to be processed at the same time / in parallel **(1)**. This improves performance when running multiple programs or programs designed for parallel processing **(1)**.

Exam tip: Mention that software must support multi-core processing to benefit. A single-threaded program will not run faster with more cores.

Worked Example 11 – Explaining cache

Question: Explain how cache size affects CPU performance. [3 marks]

Model answer:
Cache is fast memory located close to the CPU that stores frequently used data **(1)**. A larger cache can store more data, reducing the number of times the CPU must access slower main memory **(1)**. This reduces waiting time and allows the CPU to process instructions more quickly **(1)**.

Exam tip: Emphasise the speed difference between cache and RAM. The key benefit of cache is avoiding slow memory accesses.

Worked Example 12 – Comparing performance factors

Question: A school is purchasing new computers. Explain how clock speed, number of cores, and cache size affect performance. The school should consider which factor is most important for students running multiple applications simultaneously. [6 marks]

Model answer:
Clock speed: A higher clock speed means more Fetch-Execute cycles per second **(1)**. This makes individual tasks run faster **(1)**.
Number of cores: More cores allow the CPU to process multiple tasks simultaneously **(1)**. This improves performance when running multiple applications at the same time **(1)**.
Cache size: Larger cache stores more frequently used data close to the CPU **(1)**. This reduces delays when accessing data from slower RAM **(1)**.
Recommendation: For running multiple applications simultaneously, the number of cores is most important because each core can handle a different application independently.

Exam tip: In application questions, relate your answer to the specific context. Here, multi-tasking makes cores the priority.

Worked examples summary

The following table summarises the 12 worked examples covered in this topic, organised by section and exam skill.

Example	Focus	Marks	Key skill
1	Explaining the ALU	2	Describe arithmetic and logical operations
2	Explaining the Control Unit	3	Describe fetch, decode, coordination roles

Example	Focus	Marks	Key skill
3	Explaining buses	4	Compare address and data bus functions
4	Explaining registers	3	Describe purpose and characteristics
5	Describing the Fetch-Execute cycle	6	Explain all three stages in detail
6	Control Unit in the FE cycle	3	Link CU role to each stage
7	Buses in the FE cycle	3	Explain bus usage during fetch
8	Why the cycle repeats	2	Explain continuous operation
9	Clock speed and performance	2	Link speed to cycles per second
10	Multiple cores and performance	3	Explain parallel processing
11	Cache size and performance	3	Explain reduced memory access delays
12	Comparing all performance factors	6	Apply factors to a context

Quick reference

CPU components

Component	Definition (mark-scheme style)
ALU	Performs arithmetic operations (add, subtract, multiply, divide) and logical operations (comparisons, AND, OR, NOT)
Control Unit	Fetches and decodes instructions; sends control signals to coordinate CPU operations
Clock	Generates regular timing pulses to synchronise CPU operations; speed measured in Hertz
Registers	Small, very fast storage locations within the CPU; hold data and instructions currently being processed
Bus	A collection of wires through which data/signals are transmitted between components
Address bus	Carries memory addresses from CPU to memory; unidirectional

Component	Definition (mark-scheme style)
Data bus	Carries data between CPU and memory; bidirectional
Control bus	Carries control signals (read, write, clock); bidirectional

Fetch-Execute cycle stages

Stage	AQA definition
Fetch	The next instruction is fetched to the CPU from main memory
Decode	The instruction is decoded to work out what it is
Execute	The instruction is executed (carried out); this may include reading/writing from/to main memory

Performance factors

Factor	How it improves performance
Higher clock speed	More Fetch-Execute cycles per second; more instructions processed per second
More cores	Multiple instructions processed simultaneously; better for multi-tasking and parallel processing
Larger cache	More frequently used data stored close to CPU; fewer slow accesses to main memory

Units of clock speed

Unit	Value	Example
Hz (Hertz)	1 cycle per second	Original Intel 4004: 740 kHz
MHz (Megahertz)	1 million cycles per second	500 MHz
GHz (Gigahertz)	1 billion cycles per second	3.5 GHz (typical modern CPU)

Typical pitfalls

Pitfall 1 — Confusing ALU and CU roles.
Wrong: "The ALU fetches instructions from memory."
Right: The Control Unit fetches instructions. The ALU performs calculations and comparisons.

Pitfall 2 — Saying registers store programs.

Wrong: "Registers store the program being run."
Right: Registers store only the data/instruction currently being processed. Programs are stored in main memory.

Pitfall 3 — Mixing up bus directions.
Wrong: "The address bus carries data in both directions."
Right: The address bus is unidirectional (CPU to memory). The data bus is bidirectional.

Pitfall 4 — Forgetting decode stage.
Wrong: "The Fetch-Execute cycle has two stages: fetch and execute."
Right: There are three stages: fetch, decode, and execute. The decode stage is essential.

Pitfall 5 — Claiming more cores always means faster.
Wrong: "An 8-core CPU is always faster than a 4-core CPU."
Right: Software must support parallel processing to benefit from multiple cores. Single-threaded programs gain no benefit from extra cores.

Pitfall 6 — Confusing cache and registers.
Wrong: "Cache is inside the CPU and is the fastest memory."
Right: Registers are inside the CPU and are fastest. Cache is very fast but sits between the CPU and RAM in the hierarchy.

Pitfall 7 — Saying clock speed is measured in bytes.
Wrong: "Clock speed is measured in gigabytes."
Right: Clock speed is measured in Hertz (Hz, MHz, GHz). Bytes measure storage capacity, not speed.

Pitfall 8 — Incomplete FE cycle explanations.
Wrong: "In the execute stage, the CPU runs the instruction."
Right: Be specific: "The instruction is executed, which may involve the ALU performing a calculation or reading/writing from/to main memory."

Pitfall 9 — Forgetting the continuous nature.
Wrong: "The Fetch-Execute cycle runs once for each program."
Right: The cycle repeats continuously, processing each instruction in sequence while the computer runs.

Pitfall 10 — Vague performance explanations.
Wrong: "More cache makes the computer faster."

Right: Explain why: "More cache stores more frequently used data close to the CPU, reducing the number of slow accesses to main memory."

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